ECSE 331
Laboratory No. 3
MOSFET DC Characteristics in PSPICE

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Seyed Yazdan Zinati
260928077
seyed.zinati@mail.mcgill.ca

James Wu
260929944
james.wu4@mail.mcgill.ca

***Abstract*—A MOSFET is a three-terminal device that allows current to flow through it when a certain threshold voltage is reached. The three terminals of a MOSFET are called the gate, the source, and the drain. There are two types of MOSFETs: NMOS FET and PMOS FET. Current flows from the drain to the source for NMOS FET and vice versa for PMOS FET. For NMOS FET, current will not flow through the device unless the voltage across the gate terminal and the source terminal V_{GS} is greater than a certain threshold voltage V_t. For PMOS FET, current will not flow through the device unless V_{GS} is less than V_t. MOSFETs have two modes of operation: triode and saturation. The difference between these two modes is that the equation to calculate the current flowing through the MOSFET I_D is different. In triode mode, I_D increases much more rapidly, as the voltage across the drain and source increases V_{DS}, than in saturation mode. NMOS FETs operate in triode mode when $V_{DS} < V_{GS} - V_t$, and they operate in saturation mode when $VDS > V_{GS} - V_t$. For PMOS FETs, the conditions of the operation modes are the opposite of NMOS FET; triode mode when $V_{DS} > V_{GS} - V_t$ and saturation mode when $V_{DS} < V_{GS} - V_t$.**

I. Introduction

MOSFETs are three-terminal devices that allows current to flow through it when V_{GS} is either greater than or less than V_t. The operating mode of the MOSFET depends on V_{DS} compared to $V_{GS} - V_t$.

In this laboratory, we conduct experiments on a circuit that has an NMOS FET using circuit simulation software. PSPICE software is used to build and simulate circuits. The circuit that will be built using PSPICE is shown in Figure 1. We first examine the $I_D - V_{DS}$ characteristics of the MOSFET. Then, we determine the MOSFET transconductance. Finally, we examine the effects of temperature on MOSFET.

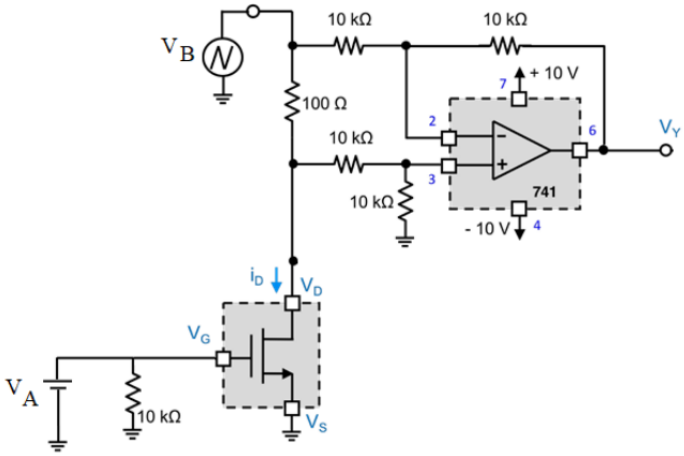


Fig. 1. Setup for measuring MOSFET $I - V$ characteristics

A. Task 1

The first part of the lab involves measuring NMOS FET characteristics. First, we plot V_{DS} vs time and I_D vs time. It should be observed that I_D remains zero despite any change in V_{DS}. We then determine the threshold voltage of the NMOS FET by increasing V_{GS} until I_D exceeds 0.5mA.

Next, we plot the voltage output (V_Y) of the op-amp divided by 100 vs time and the current through the 100Ω resistance vs time. These two plots are then compared with each other and examined.

Finally, we increase V_{GS} in increments of 1V until V_{GS} reaches 5V while plotting I_D vs V_{DS} for each value of V_{GS}. From the plot of I_D vs V_{DS} for each V_{GS} value, we can obtain the early voltage of the MOSFET. This can be done by extending the linear trends of the saturation modes of the traces to the left until they reach $I_D = 0$. The corresponding V_{DS} value is the negative value of the early effective voltage.

B. Task 2

The second part of the lab involves the computation of the MOSFET transconductance (g_m) of the NMOS FET. The transconductance is found using the following equation:

$$g_m = \frac{\Delta i_D}{\Delta v_{GS}} \tag{1}$$

The circuit must first be modified by replacing the saw-tooth signal V_B with a DC voltage source set to 5V and V_A needs to be set to 3V. I_D and V_GS are recorded at these values and they're recorded again when V_A is adjusted by 0.1V. ΔI_D and ΔV_{GS} are then found by taking the differences between the measured V_{GS} and I_D and the transconductance is computed.

C. Task 3

The third part of the lab examines the effect of temperature on the MOSFET. First, the circuit must be restored back to its original state in Figure1 by using the saw-tooth waveform as V_B and setting V_A to 3V. The simulation is conducted at temperatures 10°C, 20°C, 27°C, 35°C and 50°C. The I_D vs V_{DS} plots at these temperatures are compared.

II. Analysis

A. MOSFET I_D-V_{DS} Characteristics

Task 1 of the experiment involves examining the I-V characteristics of the MOSFET.

We first construct the circuit shown in Figure1 in PSPICE using a uA741 op-amp, a MbreakN3 NMOS FET with $W = 10\mu m$ and $L = 1\mu m$ and V_B as a saw-tooth waveform over a 0V to +10V signal range using the VPULSE component with the following characteristics:

$V1 = 0$
$V2 = 10$
$TD = 0$
$TR = 0.49e^{-3}$
$TF = 0.49e^{-3}$
$PW = 1e^{-5}$
$PER = 1e^{-3}$

We implement the -10V and +10V voltages on pin 4 and pin 7 of the uA741 op-amp using two DC voltage sources. The constructed schematic in PSPICE is shown in Figure 2.

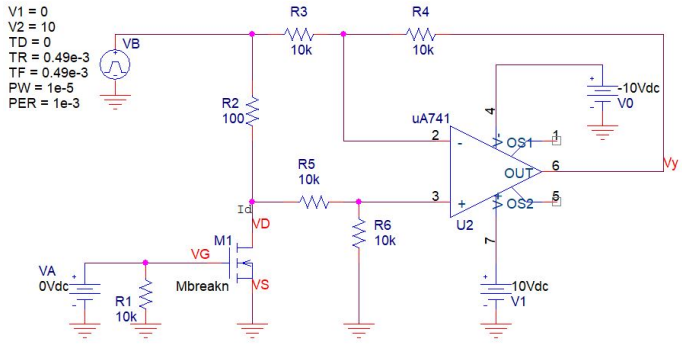


Fig. 2. Task1 circuit number 1

Initially, V_A is set to 0V. When the circuit is constructed, we perform a time-domain simulation with $V_A = 0$ to obtain plots of V_{DS} vs time and I_D vs time. The plots of V_{DS} vs time and I_D vs time can be seen in Figure 3 and Figure 4 respectively.

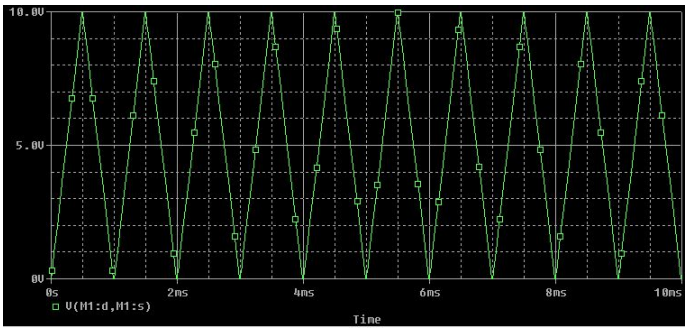


Fig. 3. V_{DS} vs time with $V_A = 0V$

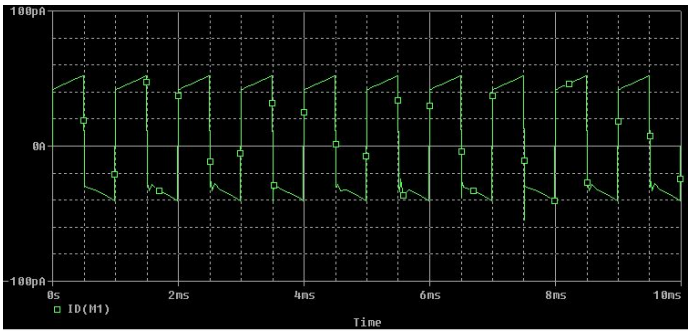


Fig. 4. I_D vs time with $V_A = 0V$

It can be seen from Figure 3 that the voltage V_{DS} follows a triangular waveform over a 0V to 10V signal range with a period of 1ms. On the other hand, the FET drain current simulation shown in Figure 4, is in the range of picoamperes which is close to 0. This is expected as the gate voltage V_G is equal to zero. Therefore, the drain current I_D remains negligible irrespective of any change in the drain-source voltage V_{DS}.

Next, to determine the threshold voltage V_t, V_{GS} is increased in 1V increments until I_D becomes greater than 0.5mA. Once I_D becomes greater than 0.5mA, the voltage V_{GS} is considered to be the threshold voltage V_t. To increase V_{GS} in 1V increments, V_A is increased in 1V increments as the value of V_A is equal to the value of V_{GS}.

The value of V_A is increased from 0V to 1V and the corresponding I_D vs time plot is obtained. At $V_{GS} = 1V$, the I_D vs time plot can be seen in Figure 5. From Figure 5, we can see that I_D isn't greater than 0.5mA, thus we need to continue increasing V_{GS} by 1V.

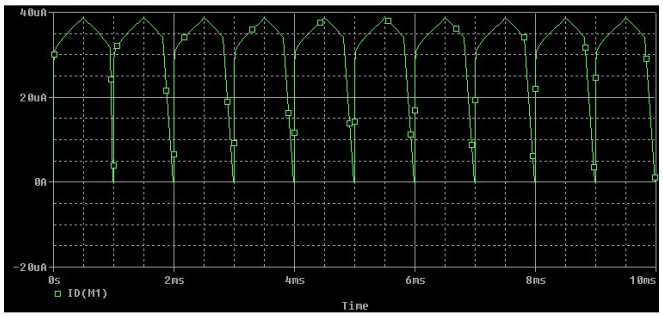


Fig. 5. I_D vs time with $V_A = 1V$

After the value of V_A is increased from 1V to 2V, its corresponding I_D vs time plot is obtained. The plot of I_D vs time can be seen in Figure 6. Based on the plot, I_D is found to be approximately 0.65mA which is greater than 0.5mA. Thus, the threshold voltage V_t is determined to be 2V.

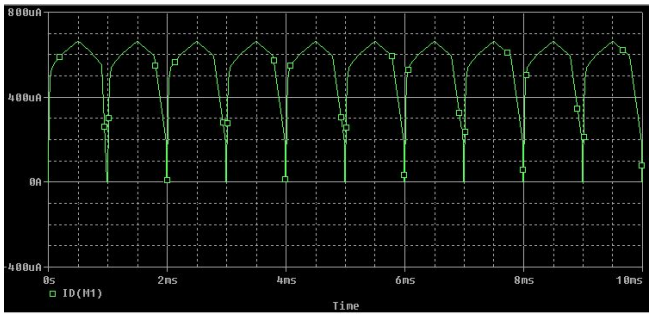


Fig. 6. I_D vs time with $V_A = 2V$

With V_A set to 2V, the output voltage of the op-amp (V_Y) is compared with the current through the 100Ω resistor when $V_A = V_t$. The plots of $\frac{V_Y}{100}$ vs time and the current through the 100Ω resistor vs time are obtained when $V_A = 2V$. These plots can be seen in Figure 7 and Figure 8.

Based on the plots, the value of $\frac{V_Y}{100}$ is the negative value of the current through the 100 Ω resistor. The relation between these two values can be better seen in Figure 9.

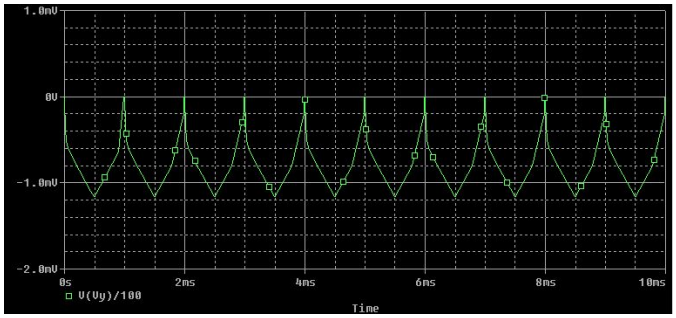


Fig. 7. $\frac{V_Y}{100}$ vs time with $V_A = 2V$

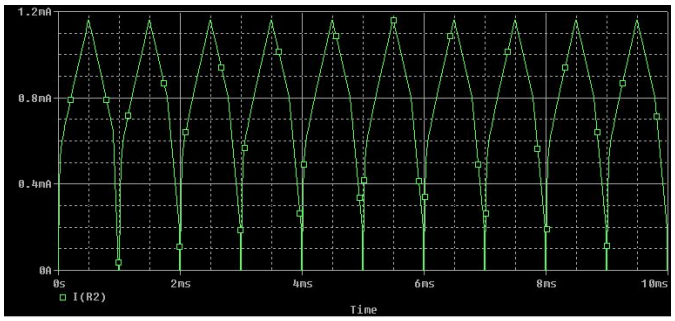


Fig. 8. I through the 100 Ω resistor vs time with $V_A = 2V$

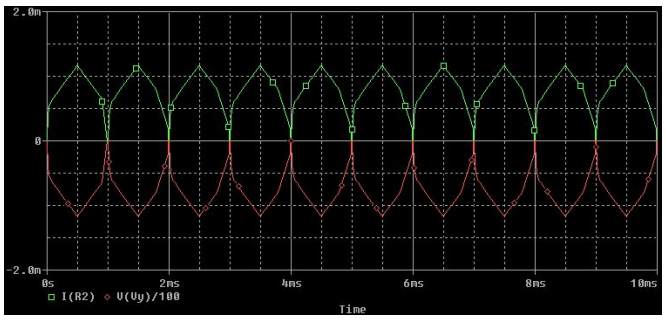


Fig. 9. I through the 100 Ω resistor vs time and $\frac{V_Y}{100}$ with $V_A = 2V$

Next, V_{GS} is increased from the threshold voltage of 2V in 1V increments until it reaches 5V. Then, the traces of I_D vs V_{DS} at each value of V_{GS} are obtained. The traces of I_D vs V_{DS} combined in a single plot can be seen in Figure 10 and 11. From the plot, the voltage region of V_{DS} at which the NMOS operates in triode mode increases as V_{GS} increases; larger V_{DS} is needed to enter saturation mode. Also, the I_D when the NMOS enters saturation mode increases when V_{GS} increases.

These observations are consistent with the theoretical equations of NMOS. NMOS operates in saturation mode when $V_{DS} > V_{GS} - V_t$; V_{DS} needs to increase when V_{GS} increases to enter saturation mode which is consistent with the observations of the plot. The value of ID when the NMOS operates in triode mode is:

$$I_D = k_n^{'}(\frac{W}{L})[(V_{GS} - V_t)V_{DS} - 0.5(V_{DS})^2]$$

and the value of I_D when operating in saturation is:

$$I_D = 0.5(k_n^{'})(\frac{W}{L})(V_{GS} - V_t)^2(1 + \lambda V_{DS})$$

Based on these equations, I_D increases when V_{GS} increases in both triode and saturation mode. Thus, the observation of I_D increasing when V_{GS} is increased is consistent with theory.

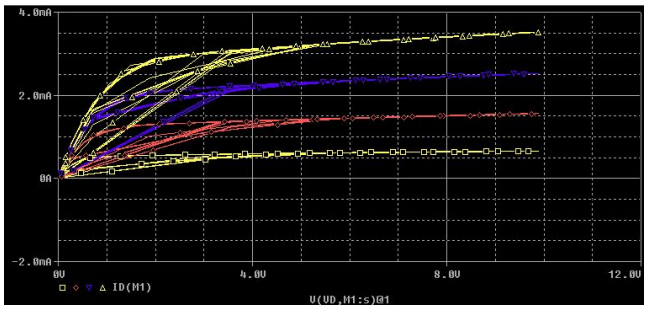


Fig. 10. I_D vs V_{DS} for V_G equal to 2V, 3V, 4V, and 5V

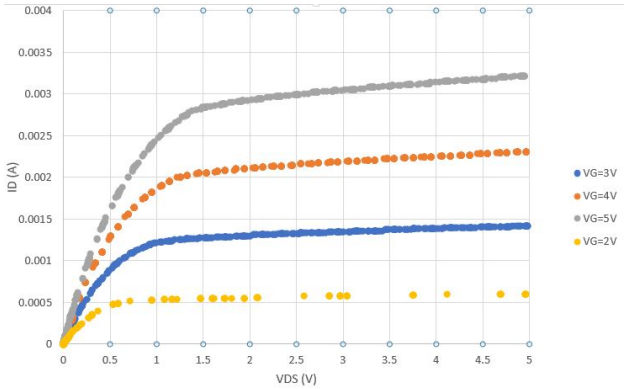


Fig. 11. I_D vs V_{DS} for V_G equal to 2V, 3V, 4V, and 5V

The early voltage for a MOSFET is 1/λ and can be found by extending the linear trendline of the I_D vs V_{DS} traces in saturation mode towards the left of the plot until they meet at $I_D = 0$. The V_{DS} value when they meet at $I_D = 0$ is -1/λ. From Figure 12, the average V_{DS} value that the extended trendlines reach zero is approximately -43V. Thus, the early voltage is determined to be approximately 43V.

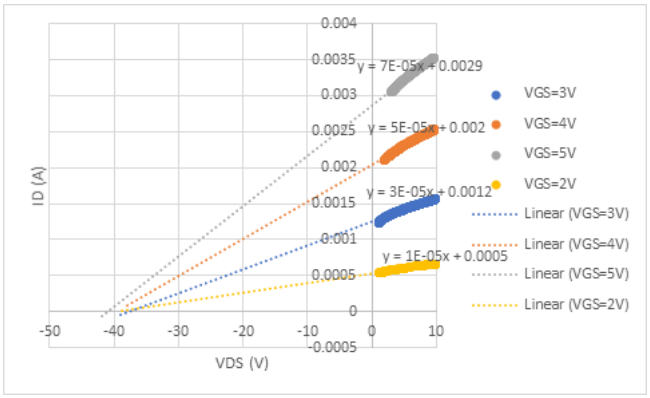


Fig. 12. I_D vs V_{DS} for V_G equal to 2V, 3V, 4V, and 5V

B. MOSFET Transconductance

Task 2 of the experiment involves computing the transconductance (g_m) of the MOSFET. The transconductance can be found using Equation 1.

First, the saw-tooth signal source V_B is replaced with a DC voltage source and set to 5V while V_A is set to 3V. V_{GS} and I_D are measured at these values of V_B and V_A. When V_A is 3V, V_{GS} is found to be 3V and I_D is found to be 1.417mA as shown in Figure 13.

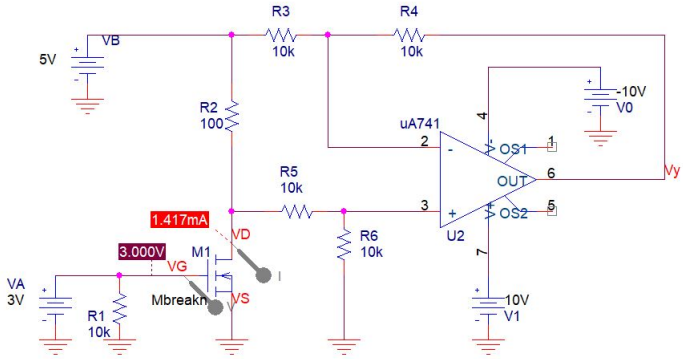


Fig. 13. I_D measurement with $V_A = 3V$ and $V_B = 5V$DC

The value of V_A is increased by 0.1V; the new value of V_A is 3.1V, thus ΔV_{GS} is 0.1V. When V_A is 3.1V, V_{GS} is found to be 3.1V and I_D is found to be 1.503mA (Figure 14). Thus ΔI_D is 0.86mA/V.

using Equation 1:

$$g_m = \frac{\Delta i_D}{\Delta v_{GS}} = \frac{1.503 - 1.417}{0.1} = 0.86^{mA}/_{V}$$

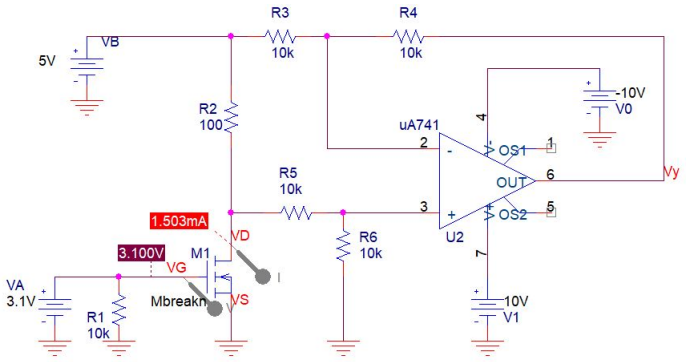


Fig. 14. I_D measurement with $V_A = 3V$ and $V_B = 5V$DC

C. MOSFET Temperature Effects

Task 3 of the experiment involves examining the effects of temperature on the $I_D - V_{DS}$ characteristics of the MOSFET. The circuit is restored to its original state by reconstructing V_B as the saw-tooth waveform from Task 1 and V_A is set to 3V. The simulation is then run at 5 different temperatures: 10°C, 20°C, 27°C, 35°C and 50°C. The traces of the $I_D - V_{DS}$ characteristics at each temperature are obtained and compared. The plot of I_D vs V_{DS} containing the traces ran at different temperatures can be seen in Figure 15.

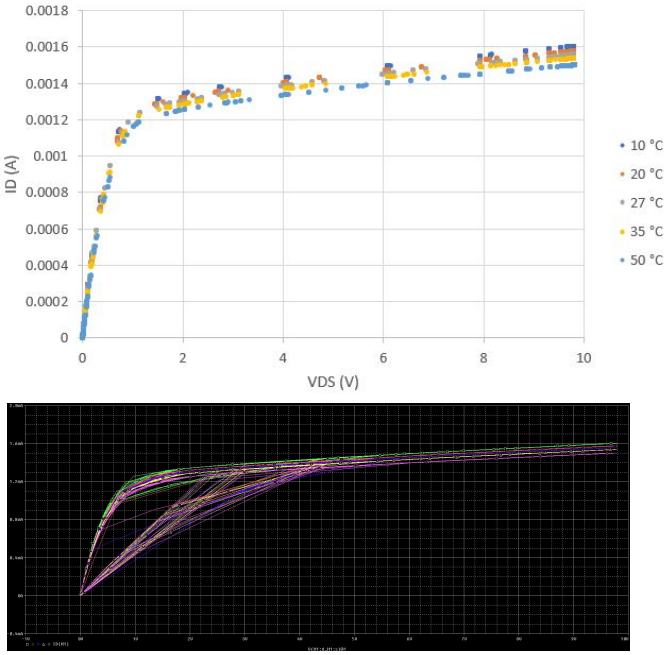


Fig. 15. $I_D - V_{DS}$ characteristics at 5 different temperatures

From the plot in Figure 15, I_D during saturation is lower for higher temperatures compared to I_D at lower temperatures. At gate voltages significantly greater than the threshold voltage, drain current decreases as temperature increases. As temperature increases, carrier mobility decreases. Thus, when the gate voltage is significantly greater than the threshold voltage, the

carrier mobility decreases which causes the drain current to be lower.

III. Conclusion

In conclusion, we learned about the DC characteristics of MOSFETs by simulating them in PSPICE. At first, we analyzed the MOSFET $I_D - V_{DS}$ characteristics through multiple experiments and plots. We found characteristics such as the NMOS threshold and the early voltage. In the second part, we ran studies to find the transconductance g_m of our device. Finally, we plot the $I_D - V_{DS}$ characteristics of our device at different temperatures to study MOSFET temperature effects. All the mentioned analysis were also verified with the studied theory.